

Nuclear Reactor Safety

AQA A-Level Physics Specification (2026)

Name: _____

Date: _____

1. Which of the following is an example of low level nuclear waste?

- A. Fission Fragments
- B. Protective Clothing
- C. Chemical Sludges
- D. Cooling Tower Vapours

2. How is low level nuclear waste disposed of?

- A. Encased in steel and buried deep underground
- B. Mixed with concrete and gravel then sent to landfill
- C. Compacted and buried on site
- D. No special disposal procedures are required

3. Which of the following is an example of intermediate level nuclear waste?

- A. Fission Fragments
- B. Protective Clothing
- C. Chemical Sludges
- D. Cooling Tower Vapours

4. How is intermediate level nuclear waste disposed of?

- A. Mixed with high level nuclear waste and both are disposed of together
- B. Encased in glass and then placed in thick steel casks to be buried deep underground
- C. Incinerated and buried deep underground
- D. Encased in concrete and placed in thick steel casks to be buried deep underground

5. Which of the following is an example of high level nuclear waste?

- A. Fission Fragments
- B. Protective Clothing
- C. Chemical Sludges
- D. Cooling Tower Vapours

6. Usable fragments of which elements are removed for recycling before disposing of high level waste?

- A. Pollonium and Uranium
- B. Plutonium and Uranium
- C. Pollonium and Plutonium
- D. Uranium only

7. Once any useful fragments have been removed from high level nuclear waste, how is the remaining waste disposed of?

- | | |
|---|--|
| A. Waste is placed in thick steel drums and encased in concrete to be stored in dedicated overground facilities | B. Waste is electroplated with aluminium then encased in titanium and buried deep underground |
| C. Waste is diluted with nonradioactive atoms and placed in thick titanium drums to be buried deep underground | D. Waste is vitrified then placed in thick steel drums encased in concrete and buried deep underground |

8. In an emergency shut-down, which component drops into the reactor core when power is disconnected?

- | | |
|--------------|-----------------|
| A. Fuel Rods | B. Control Rods |
| C. Moderator | D. Coolant |

9. Which concentration of uranium is considered to be enriched?

- | | |
|---------------|---------------|
| A. >90% U-235 | B. 3-5% U-236 |
| C. 3-5% U-235 | D. >90% U-236 |

10. The thick concrete shielding around the reactor is primarily used to protect workers against what danger?

- | | |
|--|-------------------|
| A. Alpha Radiation | B. Beta Radiation |
| C. Gamma Radiation & Relativistic Neutrons | D. Extreme Heat |

11. As a safety measure, the mass of fuel used in a reactor relates to the critical mass by what measure?

- | | |
|---|---|
| A. Mass of fuel used is slightly less than critical mass | B. Mass of fuel used is significantly less than critical mass |
| C. Mass of fuel used is slightly greater than critical mass | D. Mass of fuel used is equal to critical mass |

Answer Key

1. A
2. C
3. C
4. D

5. A
6. B
7. D
8. B

9. C
10. C
11. A